

Question 1 (January 1992)

A bird leaves its nest for a short horizontal flight along a straight line, and then returns.

Michelle models its distance, s metres, from the nest at time t seconds by

$$s = 25t - \frac{5t^2}{2}, \quad 0 \leq t \leq 10.$$

- (i) Find the value of s when $t = 2$. [1]
- (ii) Explain the restriction $0 \leq t \leq 10$. [2]
- (iii) Find the velocity of the bird at time t seconds. [1]
- (iv) What is the greatest distance of the bird from the nest? [2]
- (v) Michelle's teacher tells her that a better model would be

$$s = 10t^2 - 2t^3 + \frac{t^4}{10}.$$

Show that the two models agree about the time of the journey and the greatest distance travelled. Compare their predictions about velocity and suggest why the teacher's model is better. [4]

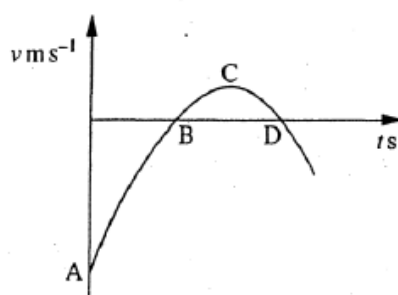
Question 2 (January 1999)

A particle is travelling on the x -axis. Its velocity $v \text{ m s}^{-1}$ is given by $v = 6(2 - t)(2t - 9)$, where t is the time in seconds.

You may wish to use the fact that $6(2 - t)(2t - 9)$ may be expressed as $78t - 12t^2 - 108$.

- (i) Find the time at which the speed of the particle in the positive direction is greatest. Calculate the speed at this time. [3]

Fig. 4 shows the velocity-time graph for the motion of the particle in the time interval $0 \leq t \leq 5$.



Not to scale

Fig. 4

- (ii) Write down the values of t and v at each of the points A, B, C and D. [2]
- At $t = 0$, the particle is 61 m from O in the positive direction.
- (iii) Calculate the position of the particle when it is first instantaneously at rest. [5]
- (iv) Determine, by considering the points where the particle is instantaneously at rest, how many times the particle passes through the origin O. [3]
- (v) Referring to the speed, direction of motion and acceleration, give a brief description of the motion of the particle between the points B and D shown in Fig. 4. [3]

Question 3 (January 1996)

1 A sports car starts from rest and moves in a straight line. Its motion is modelled as having acceleration $a \text{ m s}^{-2}$ given by

$$a = 6t \quad \text{for } 0 \leq t \leq 1,$$

$$a = 12t - 3t^2 - 3 \quad \text{for } 1 \leq t \leq 2,$$

$$a = 9 \quad \text{for } 2 \leq t \leq 4,$$

where t is the time elapsed in seconds from the start. The acceleration-time graph is shown in Fig. 1.

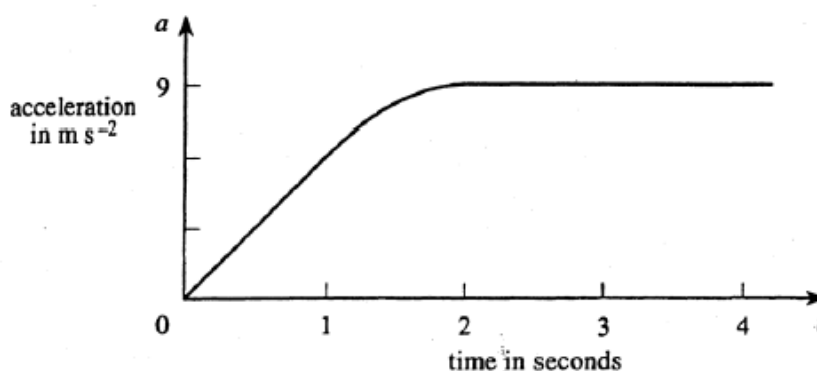


Fig. 1

(i) Find an expression for the speed of the car for $0 \leq t \leq 1$, and state the speed of the car when $t = 1$. [4]

(ii) Show that the speed of the car, v , is given by

$$v = 6t^2 - t^3 - 3t + 1 \quad \text{for } 1 \leq t \leq 2$$

and find an expression for the speed of the car for $2 \leq t \leq 4$. [6]

(iii) How far does the car travel between $t = 1$ and $t = 2$? [4]

Question 4 (June 1996)

The speed v , in metres per second, at time t seconds of a child running 100 metres is modelled over the first 4 seconds by

$$v = 4t - \frac{1}{2}t^2, \quad (0 \leq t \leq 4).$$

For $t > 4$ the child runs at a constant speed.

- (i) Find the speed of the child at times 0, 2 and 4 seconds. [1]
- (ii) Find the acceleration of the child when $t = 4$. What does this acceleration indicate about the motion of the runner? [3]
- (iii) Sketch the speed-time graph for the child's run on the grid provided. [2]
- (iv) Calculate the distance travelled by the child in the first 4 seconds. [2]
- (v) How long does it take the child to complete the 100 metres? [2]